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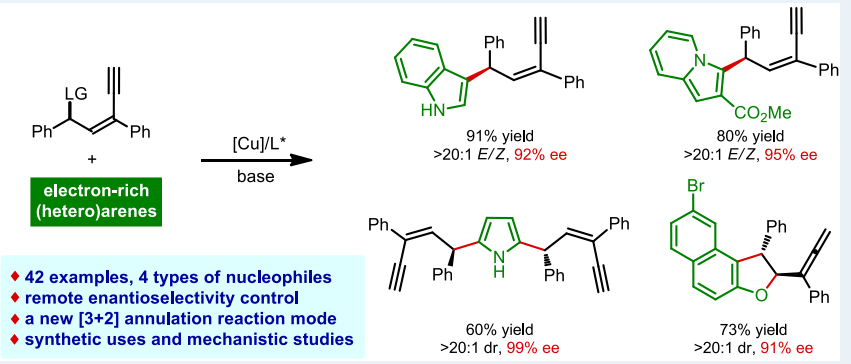
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Copper-Catalyzed Asymmetric Yne-Allylic Substitution Using Electron-Rich Arenes

Defu Luo,§ Shengtong Niu,§ Fan Gong, Chao Xu, Shouang Lan, Jinggong Liu,* Shuang Yang, and Xinqiang Fang*

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ABSTRACT: Remote stereocontrol in transition-metal catalysis is a challenging but interesting research topic. In this work, we achieved copper-catalyzed asymmetric yne-allylic substitution using electron-rich arenes and acyclic carbonates through remote enantioselectivity control. The reaction delivers a variety of enantioenriched products that contain a diverse set of valuable moieties, such as conjugated enynes, indoles, indolizines, allenes, and dihydrofurans, which are widely used in organic synthesis and act as key units in bioactive molecules and natural products. The synthetic value of this protocol has been demonstrated in a series of further transformations, and mechanistic studies have been conducted to gain more insight into the reaction.

KEYWORDS: *copper catalysis, yne-allylic substitution, 1,3-enyne, electron-rich arene, remote enantioselectivity control*

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INTRODUCTION

The asymmetric construction of C−C bonds through remote stereocontrol is a challenging but fascinating synthetic strategy because it overcomes the limits of stereorecognition of remote prochiral centers and has long been an object of great interest in asymmetric synthesis. However, such a strategy is still underdeveloped, especially in the field of transition-metal catalysis.[1] In this situation, the Ru or Cu-catalyzed asymmetric propargylic substitution has been established as a powerful tactic for the formation of stereogenic centers having alkynyl groups (Scheme 1a).[2] Recently, we disclosed the Cu-catalyzed yne-allylic substitution using various stabilized nucleophiles (Scheme 1b).[3] The key intermediate is a copper acetylide-bonded allylic cation with a copper vinyl allenylidene species as its resonance structure. The work systematically disclosed four reaction modes between a diverse set of nucleophiles and alkynylallylic carbonates and provided a blueprint for further investigations. A subsequent mission is to develop the asymmetric version of yne-allylic substitution, but it is a much more challenging task because, besides the remote stereocontrol to produce the desired α-substituted 1,3-*E*-enyne product with a high level of enantioselectivity, two other possible products, i.e., γ-substituted 1,4-enyne products and α-substituted 1,3-*Z*-enyne products are also amenable to be formed (Scheme 1b). Very recently, Lin and He reported the Cu-catalyzed asymmetric alkynylallylic substitution using amines as the nucleophiles, providing a series of 1,4-enynes bearing amino functional groups (Scheme 1c).[4a] Soon after that, Xu and Qi achieved the asymmetric Cu-catalyzed [4 + 1] annulation using yne-allylic esters and cyclic 1,3-dicarbonyls. The products are spirocyclic compounds (Scheme 1c).[4b] However, asymmetric yne-allylic substitution affording 1,3-enynes via remote stereocontrol has not been achieved.[5] 1,3-enynes are key moieties in a myriad of natural products and bioactive molecules,[6] and are also valuable building blocks that have been extensively applied in various transformations affording multiply substituted arenes, functionalized allenes,

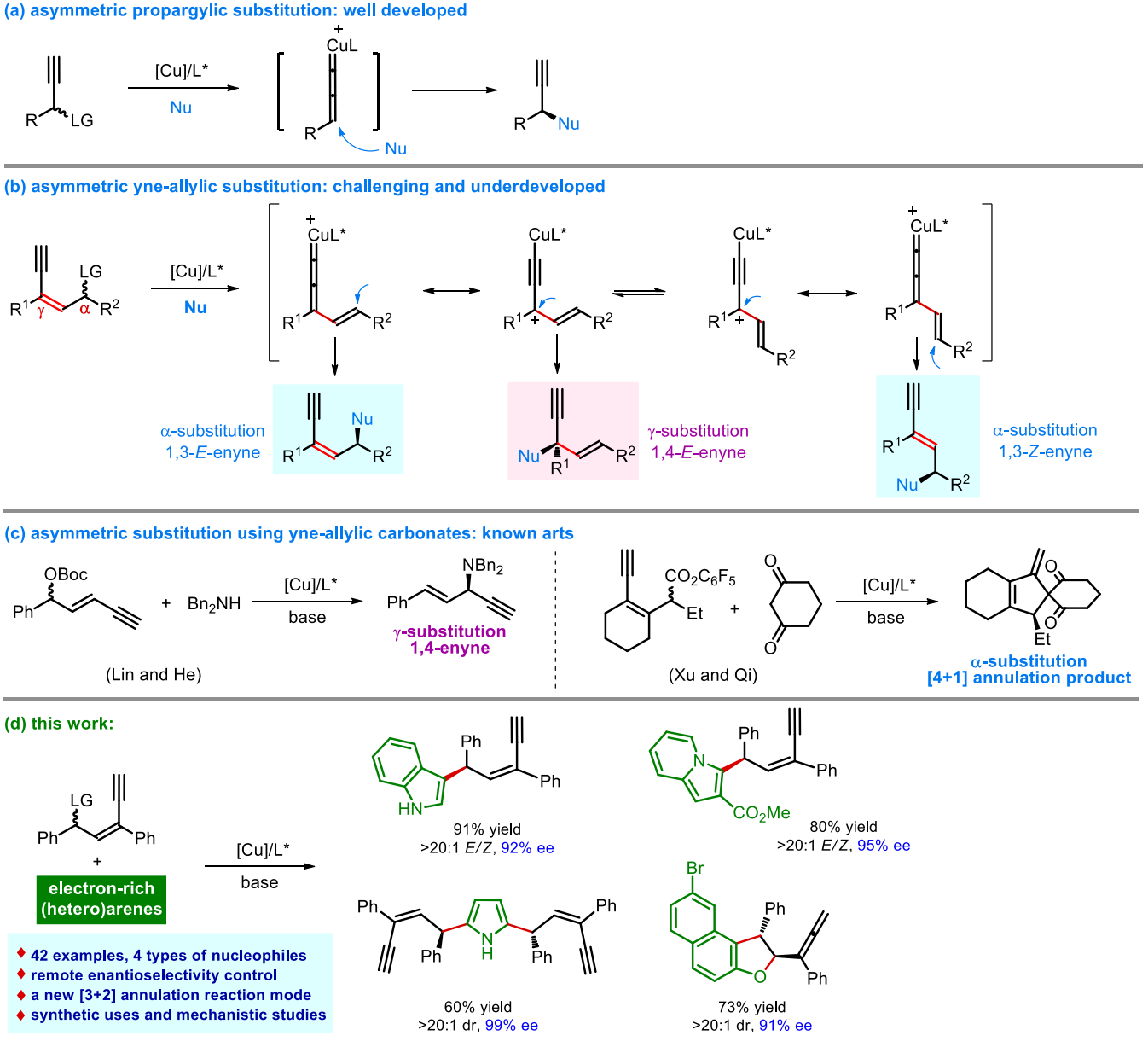
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Scheme 1. Research Background[a]



[a](a) asymmetric propargylic substitution. (b) Challenges in achieving asymmetric yne-allylic substitution. (c) Known arts of asymmetric yne-allylic substitution. (d) This work: asymmetric yne-allylic substitution using electron-rich arenes.

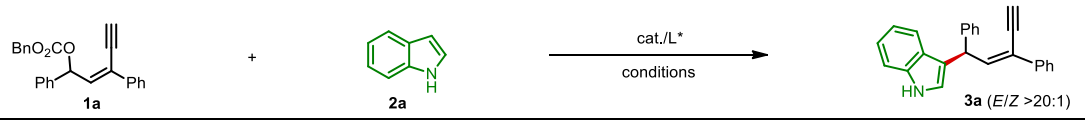
dienes, heterocycles, and other fine chemicals,[7] but protocols providing chiral 1,3-enynes bearing adjacent stereocenters still remain scarce.[8] Considering the great importance of 1,3-enynes, in this work, we describe the first highly regio- and stereoselective yne-allylic substitution using linear carbonates and electron-rich arene nucleophiles such as indoles, indolizines, pyrrols, and 2-naphthols (Scheme 1d). The work furnishes a variety of 1,3-enynes with heterocyclic moieties and dihydronaphthofuran products with allene units. Three reaction modes, i.e., monosubstitution, double substitution, and an unprecedented [3 + 2] annulation, were disclosed. It is noteworthy that a large number of drugs and pharmaceutically active compounds contain heterocyclic rings such as indoles and indolizines,[9] and allene structures widely exist in naturally occurring substances and are used in organic synthesis.[10] Thus, this work provides a convenient approach to make the above molecules. Herein, we report the results.

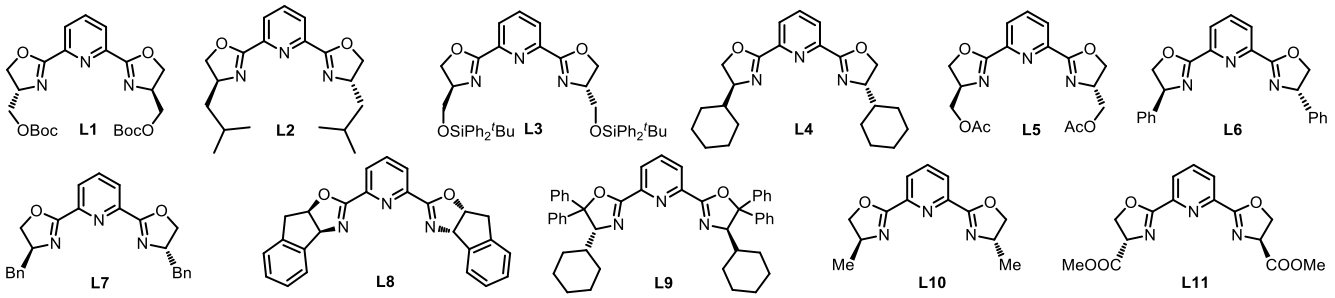
RESULTS AND DISCUSSION

We commenced by selecting readily available yne-allylic carbonate **1a** as the model substrate and indole **2a** as the nucleophile (Table 1). Our initial test of the reaction using CuI, ligand **L1**, Et_3N, and toluene at −20 °C afforded **3a** in excellent 93% yield with a promising 56% ee (Table 1, entry 1). Then, we screened more copper salts (Table 1, entries 2−7) and found that CuOTf and $Cu(OTf)_2$ both gave good outcomes, with the latter slightly better (Table 1, entries 6−7). Next, we surveyed a series of Pybox ligands **L2**−**L11** (Table 1, entries 8−17). While **L2** and **L3** performed well (Table 1, entries 8−9), **L4** was proven to be efficient in producing **3a** in excellent yield with 90% ee (Table 1, entry 10). Ligands **L5**−

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Table 1. Optimization of Reaction Conditions[a]





entry	cat (mol %)	ligand (mol %)	base (equiv)	temp (°C)	solvent (M)	yield (%)	ee (%)
1	CuI (10)	L_1 (12)	Et_3N (1.2)	−20	toluene (0.1)	93	−56
2	CuBr (10)	L_1 (12)	Et_3N (1.2)	15	toluene (0.1)	90	−39
3	$Cu(MeCN)_4PF_6$ (10)	L_1 (12)	Et_3N (1.2)	15	toluene (0.1)	92	−29
4	CuTc (10)	L_1 (12)	Et_3N (1.2)	15	toluene (0.1)	<5	0
5	$CuCF_3(PPh_3)_4$ (10)	L_1 (12)	Et_3N (1.2)	15	toluene (0.1)	94	−2
6	CuOTf (10)	L_1 (12)	Et_3N (1.2)	15	toluene (0.1)	92	−88
7	$Cu(OTf)_2$ (10)	L_1 (12)	Et_3N (1.2)	15	toluene (0.1)	93	−89
8	$Cu(OTf)_2$ (10)	L_2 (12)	Et_3N (1.2)	15	toluene (0.1)	92	87
9	$Cu(OTf)_2$ (10)	L_3 (12)	Et_3N (1.2)	15	toluene (0.1)	91	78
10	$Cu(OTf)_2$ (10)	L_4 (12)	Et_3N (1.2)	15	toluene (0.1)	92	90
11	$Cu(OTf)_2$ (10)	L_5 (12)	Et_3N (1.2)	15	toluene (0.1)	55	60
12	$Cu(OTf)_2$ (10)	L_6 (12)	Et_3N (1.2)	15	toluene (0.1)	88	85
13	$Cu(OTf)_2$ (10)	L_7 (12)	Et_3N (1.2)	15	toluene (0.1)	60	72
14	$Cu(OTf)_2$ (10)	L_8 (12)	Et_3N (1.2)	15	toluene (0.1)	75	56
15	$Cu(OTf)_2$ (10)	L_9 (12)	Et_3N (1.2)	15	toluene (0.1)	94	−86
16	$Cu(OTf)_2$ (10)	L_{10} (12)	Et_3N (1.2)	15	toluene (0.1)	93	88
17	$Cu(OTf)_2$ (10)	L_{11} (12)	Et_3N (1.2)	15	toluene (0.1)	<10	
18	$Cu(OTf)_2$ (10)	L_4 (12)	NMM (1.2)	15	toluene (0.1)	90	73
19	$Cu(OTf)_2$ (10)	L_4 (12)	$^iPr_2NEt$ (1.2)	15	toluene (0.1)	90	64
20	$Cu(OTf)_2$ (10)	L_4 (12)	Me_2NEt (1.2)	15	toluene (0.1)	93	86
21	$Cu(OTf)_2$ (10)	L_4 (12)	Et_2NMe (1.2)	15	toluene (0.1)	93	90
22	$Cu(OTf)_2$ (10)	L_4 (12)	Et_2NMe (1.2)	25	toluene (0.1)	91	88
23	$Cu(OTf)_2$ (10)	L_4 (12)	Et_2NMe (1.2)	10	toluene (0.1)	60	90
24	$Cu(OTf)_2$ (10)	L_4 (12)	Et_2NMe (1.2)	0	toluene (0.1)	trace	
25	$Cu(OTf)_2$ (10)	L_4 (12)	Et_2NMe (1.2)	15	toluene (0.05)	90	91
26	$Cu(OTf)_2$ (10)	L_4 (12)	Et_2NMe (1.2)	15	toluene (0.03)	91	92
27	$Cu(OTf)_2$ (10)	L_4 (12)	Et_2NMe (1.2)	15	PhCl (0.03)	68	82
28	$Cu(OTf)_2$ (10)	L_4 (12)	Et_2NMe (1.2)	15	THF (0.03)	60	91
29	$Cu(OTf)_2$ (10)	L_4 (12)	Et_2NMe (1.2)	15	CH_2Cl_2 (0.03)	75	56
30	$Cu(OTf)_2$ (10)	L_4 (12)	Et_2NMe (1.2)	15	dioxane (0.03)	90	92
31	$Cu(OTf)_2$ (5)	L_4 (6)	Et_2NMe (1.2)	15	toluene (0.03)	89	88

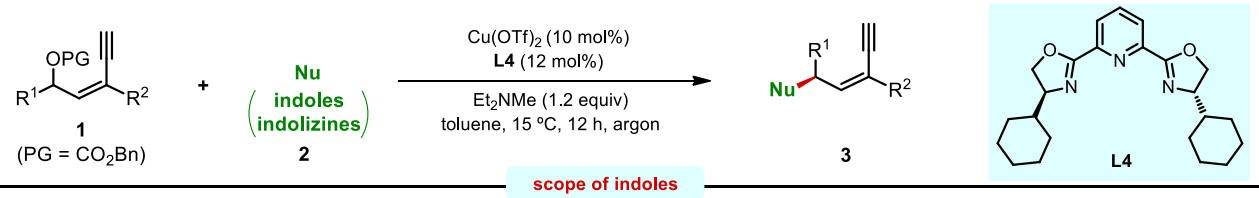
[a]Reaction conditions: **1a** (0.10 mmol), **2a** (0.12 mmol), 12 h, under argon atmosphere. All isolated yields were based on **1a**. All ee values were determined via HPLC analysis on a chiral stationary phase. All *E*/*Z* ratios were determined via ^1H NMR analysis on the reaction mixtures. NMM = *N*-Methylmorpholine.

L10 displayed moderate to good levels of enantioselectivities (Table 1, entries 11−16), and **L11** was an ineffective ligand (Table 1, entry 17). Using $Cu(OTf)_2$ and **L4**, we further evaluated a range of bases (Table 1, entries 18−21), and the reaction using Et_2NMe furnished excellent results (Table 1, entry 21). The study on the influence of the temperature on the reaction indicated that 15 °C is still the best choice (Table 1, entries 22−24). Lower concentrations of the reaction could improve the ee of the product (Table 1, entries 25−26), and the survey of other solvents (Table 1, entries 27−30) showed that dioxane also performed well (Table 1, entry 30). Finally, we tested the reaction with lower catalyst loading, and slightly lower yield and ee were obtained (Table 1, entry 31).

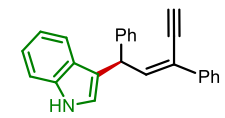
With the optimal conditions having been established, we then started to evaluate the scope and limitations of indole-mediated asymmetric substitution (Scheme 2). The introduction of electron-rich Me and OMe groups into the indole ring showed no apparent influence on the reaction, delivering **3b** and **3c** in good yields with excellent ee values (Scheme 2, **3b-c**). The use of indoles having electron-withdrawing groups such as Br and Cl slightly decreased the yields of the products, but a good level of enantioselectivity was still obtained

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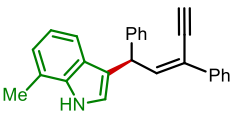
Scheme 2. Yne-Allylic Substitution Using Indoles and Indolizines[a,b]



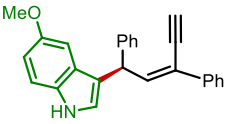
scope of indoles



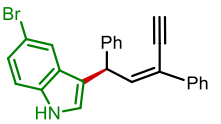
3a, 91% yield
>20:1 *E*/*Z*, 92% ee



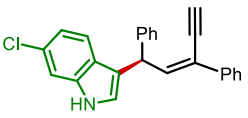
3b, 82% yield[b]
>20:1 *E*/*Z*, 94% ee



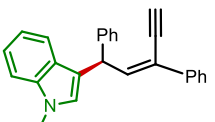
3c, 73% yield[b]
>20:1 *E*/*Z*, 90% ee



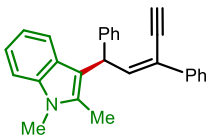
3d, 71% yield[b]
>20:1 *E*/*Z*, 87% ee



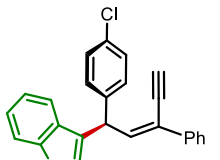
3e, 55% yield[b]
>20:1 *E*/*Z*, 85% ee



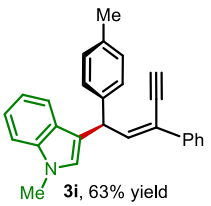
3f, 92% yield
>20:1 *E*/*Z*, 96% ee



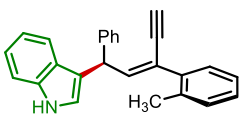
3g, 84% yield
>20:1 *E*/*Z*, 84% ee



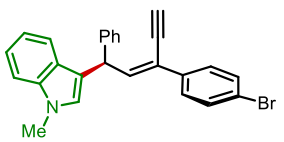
3h, 86% yield
>20:1 *E*/*Z*, 95% ee



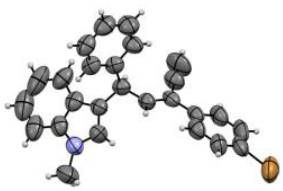
3i, 63% yield
>20:1 *E*/*Z*, 95% ee



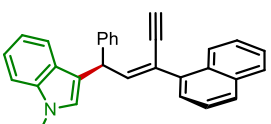
3j, 40% yield
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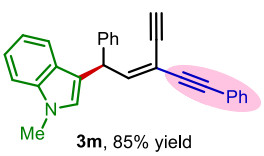
3k,74% yield
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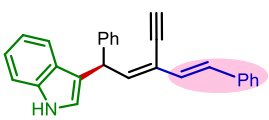
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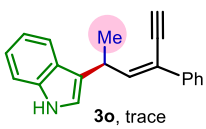
3l,82% yield
>20:1 *E*/*Z*, 91% ee



3m, 85% yield
>20:1 *E*/*Z*, 98% ee

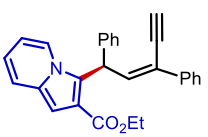


3n, 86% yield
>20:1 *E*/*Z*, 82% ee

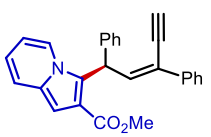


3o, trace

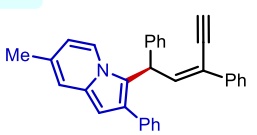
scope of indolizines



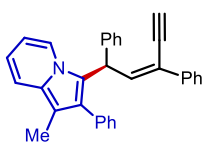
5a, 79% yield
>20:1 *E*/*Z*, 96% ee



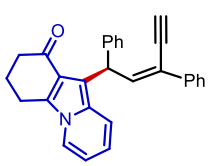
5b, 80% yield
>20:1 *E*/*Z*, 95% ee



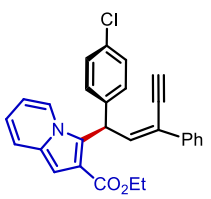
5c, 83% yield
>20:1 *E*/*Z*, 91% ee



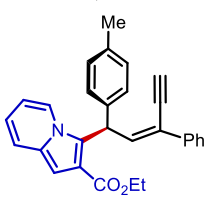
5d, 90% yield
>20:1 *E*/*Z*, 89% ee



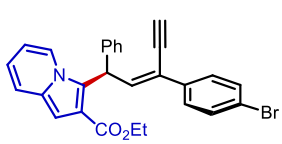
5e, 86% yield
>20:1 *E*/*Z*, 90% ee



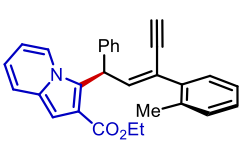
5f, 66% yield
>20:1 *E*/*Z*, 94% ee



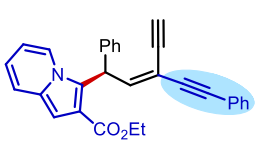
5g, 67% yield
>20:1 *E*/*Z*, 94% ee



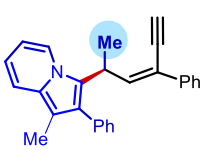
5h, 46% yield
>20:1 *E*/*Z*, 96% ee



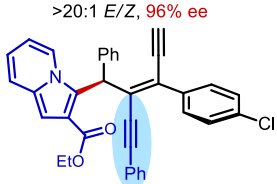
5i, 70% yield
>20:1 *E*/*Z*, 97% ee



5j, 49% yield
>20:1 *E*/*Z*, 98% ee



5k, 72% yield
>20:1 *E*/*Z*, 81% ee

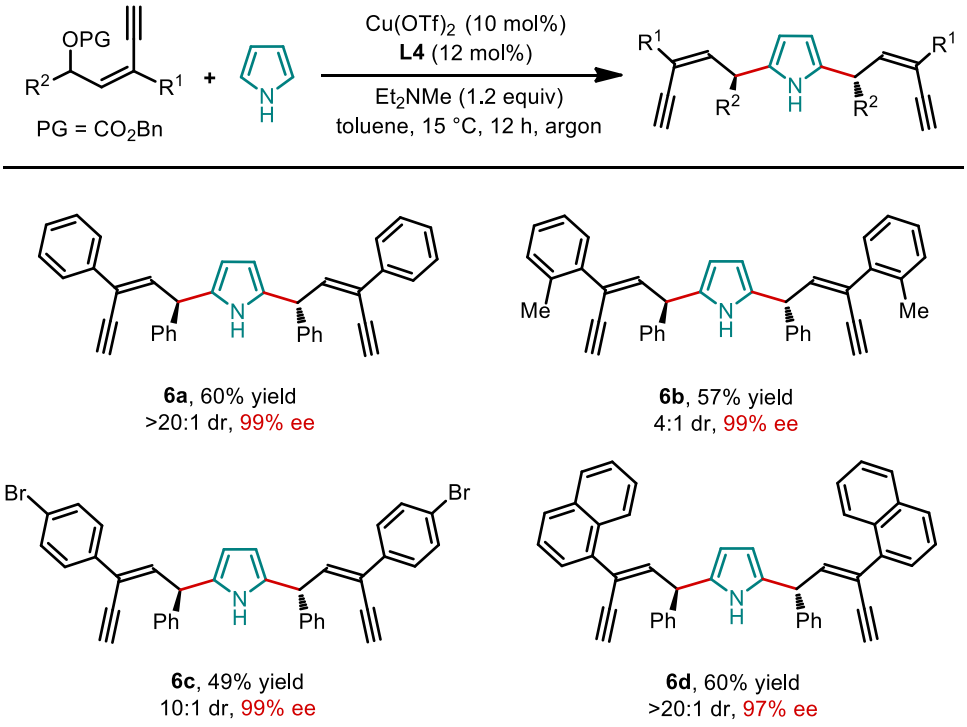


5l, 30% yield[b]
>20:1 *E*/*Z*, 75% ee

[a]All reactions were run on a 0.10 mmol scale; indoles/indolizine (1.2 equiv) was used under an argon atmosphere. All yields were of isolated products; *E*/*Z* ratios were determined via ^{1}H NMR analysis of the reaction mixtures; ee values were determined via HPLC analysis on a chiral stationary phase. [b]Reaction conditions: $Cu(OTf)_2$ (20 mol %), **L4** (24 mol %), 15 °C, 12 h.

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Scheme 3. Asymmetric Yne-Allylic Substitution Using Pyrrole[a]



[a]All reactions were run on a 0.10 mmol scale; pyrrole (1.2 equiv) was used under an argon atmosphere. All yields were of isolated products; *E*/*Z* ratios were determined via ^{1}H NMR analysis of the reaction mixtures; ee values were determined via HPLC analysis on a chiral stationary phase.

(Scheme 2, **3d**–**e**). *N*-Me indole underwent the reaction smoothly to provide **3f** in 92% yield with 96% ee (Scheme 2, **3f**), and 1,2-dimethylindole was also a suitable nucleophile (Scheme 2, **3g**). We next examined the scope of yne-allylic carbonates and found that the R^1 group could be changed to phenyl rings having different electronic properties, and both **3h** and **3i** were obtained with excellent 95% ee (Scheme 2, **3h**–**3i**). Moreover, the variation of the R^2 group with an electron-rich or electron-poor aryl ring was also possible, providing **3j** and **3k** with 94 and 96% ee, respectively (Scheme 2, **3j**–**3k**). The absolute configuration of **3k** was determined via single-crystal X-ray diffraction (SCXRD) analysis. To our delight, the R^2 group in the substrates could be replaced by naphthyl, phenylethynyl, and styryl groups (Scheme 2, **3l**–**3n**). Unfortunately, when R^1 was an alkyl group such as Me, only a trace amount of the product was detected (Scheme 2, **3o**). We also tested the reaction of 3-methylindole with **1a** but did not observe any product, and the attempts to make carbonates bearing a *Z*-olefin unit failed because the current synthetic route produces *E*-substrate exclusively.

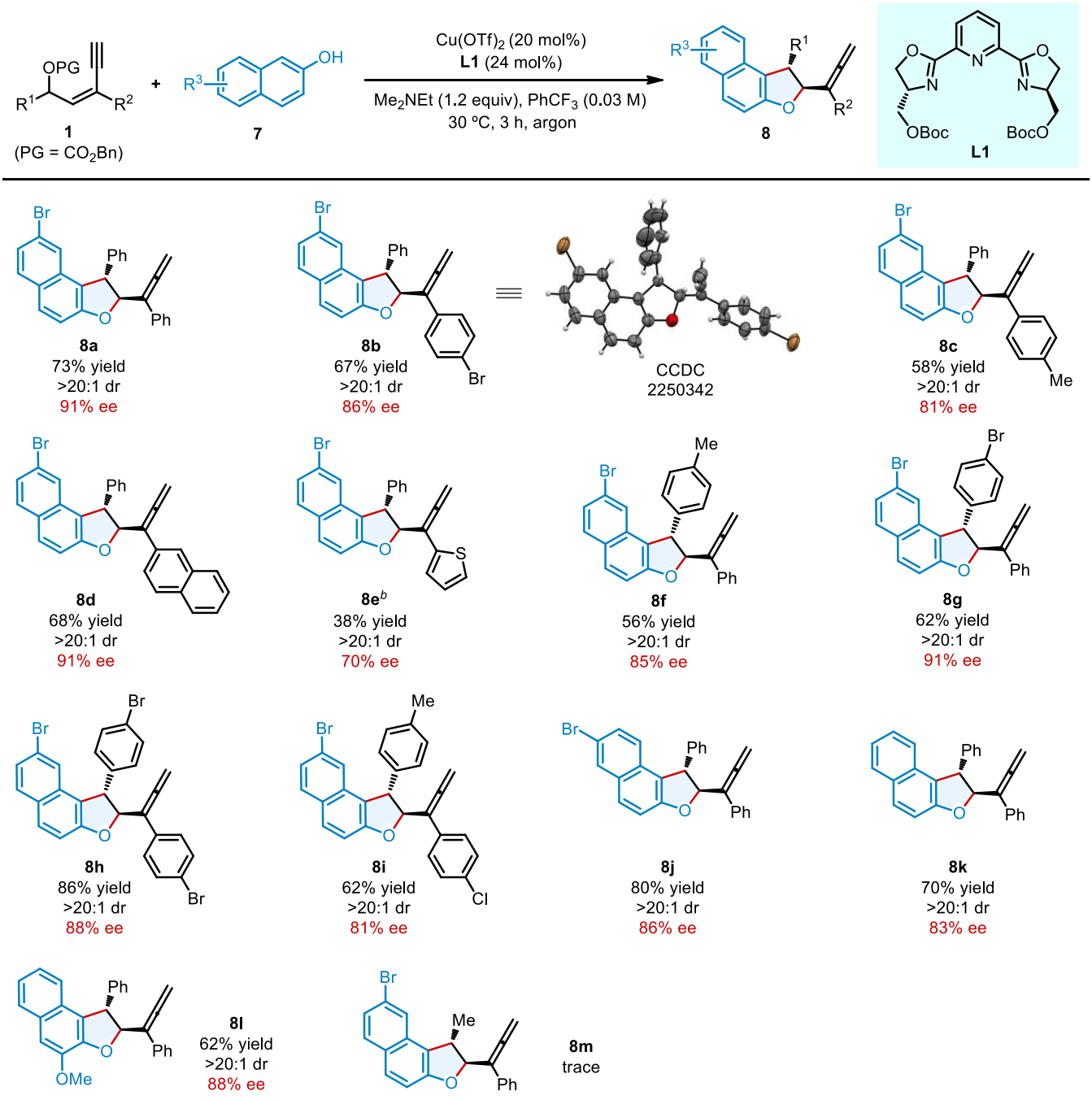
Indolizines have been found as the key substructures in many natural products and bioactive molecules, such as lamellarin D, lamellarin N, indoxam, and butoprozine, and promising anticancer, antidiabetic, and inhibitory activities have been observed in artificial indolizine derivatives.[11] Therefore, we envisioned the possibility of employing indolizines to yne-allylic substitution. Under the same conditions used for indoles, we obtained the corresponding product **5a** in 79% yield with 96% ee (Scheme 2, **5a**). Methyl ester-substituted indolizine was compatible with the reaction (Scheme 2, **5b**), and the reaction also tolerated Me/Ph-substituted indolizines (Scheme 2, **5c**–**5d**). Furthermore, the reaction using indolizine bearing a cyclic ketone moiety was also proven to be successful, releasing **5e** in a high yield with an excellent ee value (Scheme 2, **5e**). Then, we found that both R^1 and R^2 groups could be aryl rings with electron-donating and electron-withdrawing groups, and the reactions afforded **5f**–**5i** with 94–97% ee (Scheme 2, **5f**–**5i**). Additionally, the alkyne group at the R^2 position showed a limited effect on the reaction (Scheme 2, **5j**), and the use of Me at the R^1 position also proved to be successful (Scheme 2, **5k**). A comparison of the products of **3o** and **5k** shows that there might exist a match/mismatch effect between Me-substituted yne-allylic carbonate and indole or indolizine. Finally, the carbonate having a tetrasubstituted alkene unit could also deliver the corresponding product, albeit in a low yield (Scheme 2, **5l**).

The third mission of our study is to check the use of pyrrole as the nucleophilic reagent. Pyrrole has been less employed in the asymmetric allylic substitutions,[12] and only one asymmetric propargylic substitution using pyrrole has been reported by the Nishibayashi group,[2l] to the best of our knowledge. The regioselectivity between *N*-substitution and *C*-substitution, when pyrrole is used, is also an issue to be addressed. To our pleasure, we found that double yne-allylic substitution occurred, and the corresponding products were all obtained with excellent 97–99% ee (Scheme 3, **6a**–**6d**).

Having successfully achieved the asymmetric reactions using a series of electron-rich heterocyclic arenes, we further evaluated the fourth type of aromatic compound, i.e., 2-naphthalenol (Scheme 4). To our surprise, a new reaction mode that has been not disclosed previously was found, and

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Scheme 4. Asymmetric Yne-Allylic Substitution Using 2-Naphthalenols[a,b]

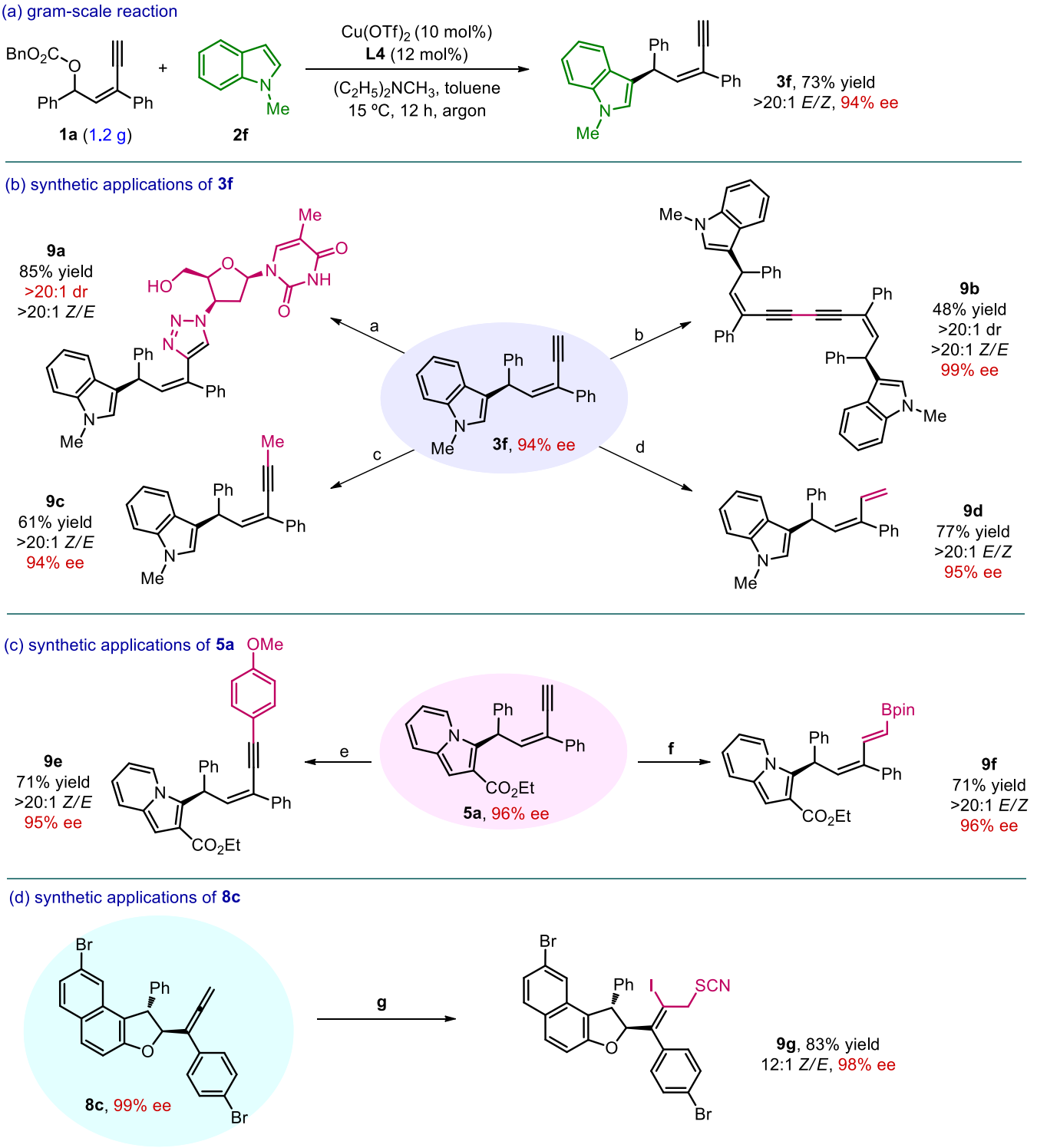


[a]All reactions were run on a 0.10 mmol scale; naphthol (1.2 equiv) was used under an argon atmosphere. All yields were of isolated products; dr values were determined via ^{1}H NMR analysis of the reaction mixtures; ee values were determined via HPLC analysis on a chiral stationary phase. [b]PG = Ac.

the reaction furnished a range of allenyl dihydronaphthofuran products via a formal [3 + 2] annulation (Scheme 4, **8a**). Allenes have been well-recognized as highly valuable synthons in downstream synthesis;[13] moreover, dihydronaphthofuran are widely found in bioactive molecules and natural products such as cassigarol D, loddigesiinol I, polynapstilbene A, and chamaecyformosanin A, but concise methods making such key structures are still underdeveloped.[14] Therefore, our protocol provides a suitable solution to this long-term challenge. We observed that the reaction tolerated carbonates bearing various aromatic R^1 and R^2 groups. For example, the R^2 group could be 4-BrC_6H_4, 4-MeC_6H_4, 2-naphthyl, and 2-thienyl, and the corresponding products were obtained in moderate to good yields with good to excellent ee (Scheme 4, **8b**−**8e**). Additionally, the R^1 group can be phenyl rings bearing either electron-donating or withdrawing substituents, delivering both **8f** and **8g** with high to excellent enantioselectivities (Scheme 4, **8f** and **8g**). Moreover, the simultaneous variation of both R^1 and R^2 groups also proved successful, releasing **8h** and **8i** smoothly (Scheme 4, **8h** and **8i**). Finally, we found that differently substituted 2-naphthols were also compatible under the reaction conditions, and the reactions afforded **8j**−**8l** in

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Scheme 5. Synthetic Applications[a]

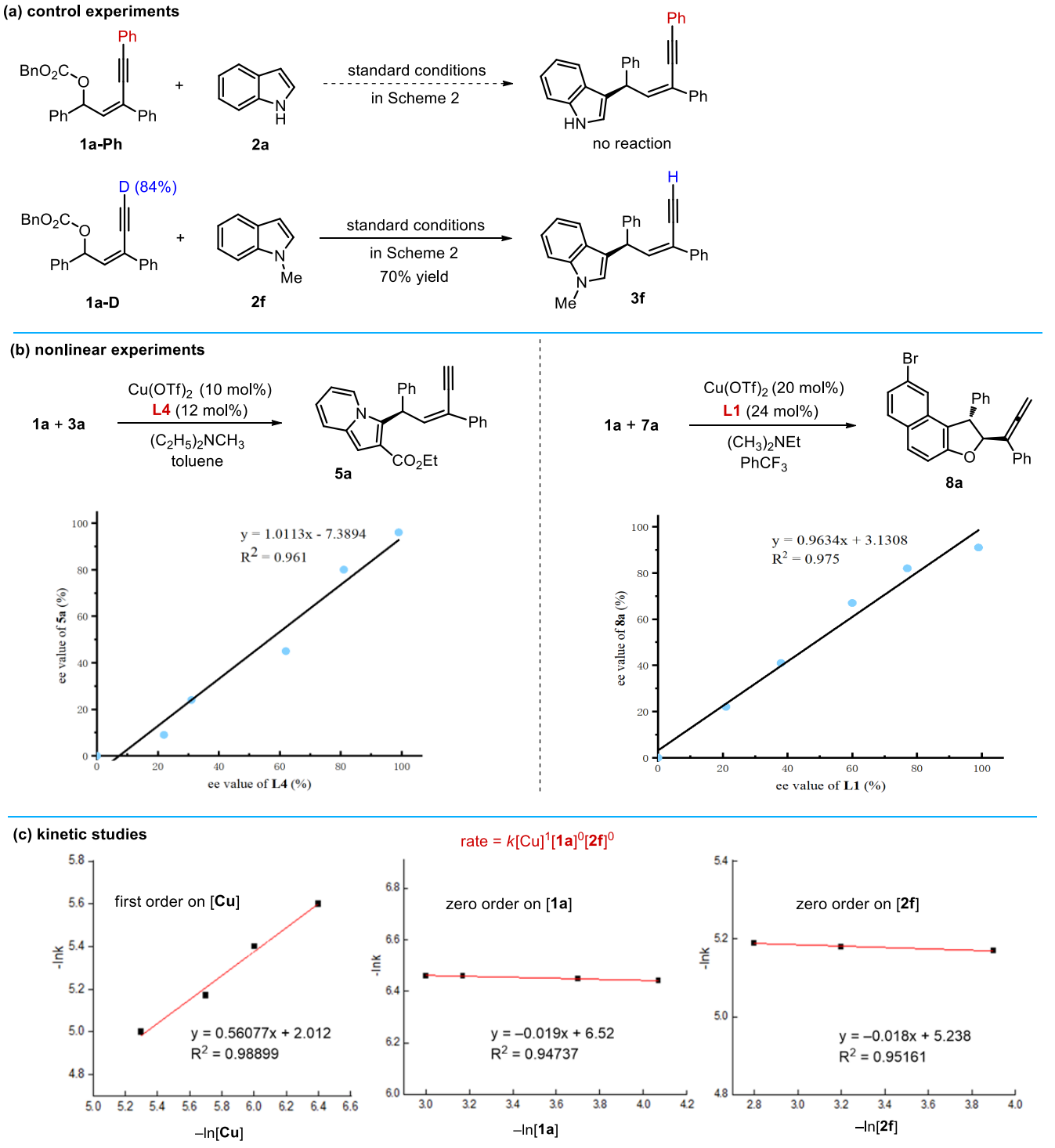


[a](a) Gram-scale reaction. (b) Synthetic applications of **3f**. (c) Synthetic applications of **5a**. (d) Synthetic applications of **8c**. Conditions: (a) Zidovudine (1.5 equiv), CuI (10 mol %), THF (0.2 M), argon, 50 °C, 24 h. (b) N,N,N′,N′-tetramethylethylenediamine (TMEDA, 1.2 equiv), CuI (10 mol %), toluene (0.2 M), 25 °C, 5 h, argon. (c) ^{n}BuLi (3 equiv), MeI (3 equiv), THF (0.2 M), −78−20 °C, 3 h, argon. (d) 10% Lindlar catalyst (10 mg), EtOAc (0.2 M), H_2, 20 °C, 12 h. (e) 4-Iodoanisole (1 equiv), Et_3N (3 equiv), CuI (10 mol %), $Pd(PPh_3)_2Cl_2$ (5 mol %), THF (0.2 M), 30 °C, 8 h, argon. (f) B_2Pin_2 (2 equiv), CuI (10 mol %), NaO^{t}Bu (20 mol %), EtOH (0.2 M), 25 °C, 11 h, argon. (g) KSCN (3.0 equiv), I_2 (2.0 equiv), 1,4-dioxane, 65 °C, 18 h.

good yields with high ee values (Scheme 4, **8j**−**8l**). Unfortunately, the carbonate having an aliphatic methyl group failed to undergo the reaction, and no product was observed (Scheme 4, **8m**). We have also attempted to isolate or detect the possible intermediate product before the

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Scheme 6. Mechanistic Studies[a]



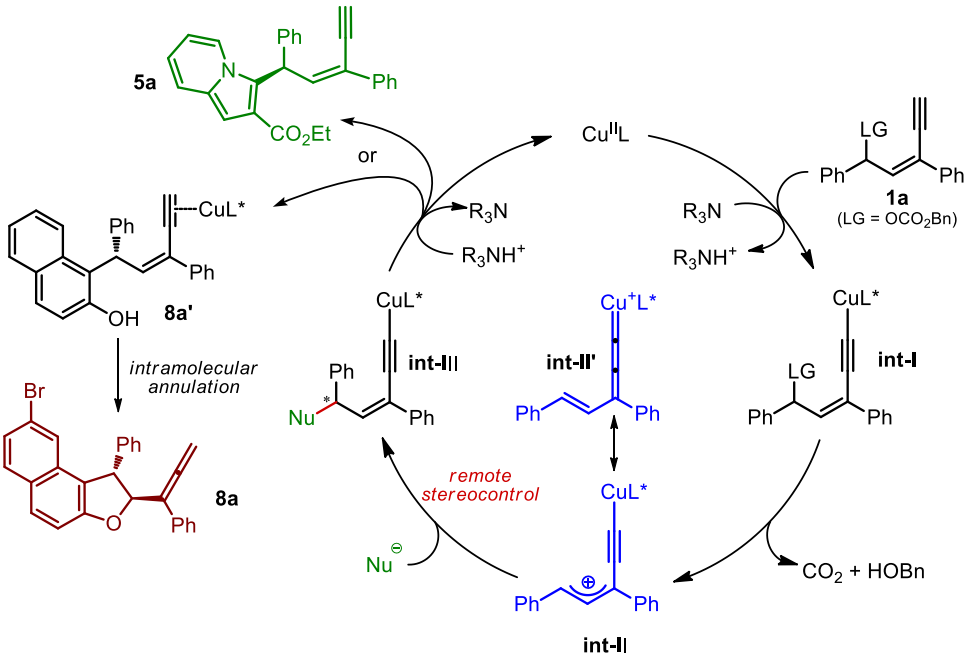
[a](a) Control experiments. (b) Nonlinear experiments. (c) Kinetic studies.

cyclization, but all failed. The second step of allenyl dihydronaphthofuran formation proceeds very quickly.

The gram-scale reaction using 1.2 g of **1a** proceeded smoothly to produce **3i** in 73% yield with 94% ee (Scheme 5a). Besides this, the synthetic value of this protocol was witnessed by a range of further transformations of the products. For instance, the alkyne unit of **3f** could undergo click reaction with zidovudine to afford **9a** in high yield with excellent dr (Scheme 5b, **9a**), and the dimerization of **3f** was achieved under copper catalysis, producing the corresponding product with 99% ee (Scheme 5b, **9b**). The terminal alkyne moiety was able to be methylated without erosion of the ee (Scheme 5b, **9c**), and the regioselective hydrogenation of **3f** allowed access to diene **9d** (Scheme 5b, **9d**). In addition, **5a** could undergo Sonogashira coupling or protoboryl addition to produce **9e** and **9f**, respectively (Scheme 5b, **9e** and **9f**). Moreover, the allene moiety of **8c** could be difunctionalized to generate **9g** efficiently under known conditions (Scheme 5c, **9g**).[15]

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Scheme 7. Proposed Mechanisms



Further studies were conducted to obtain more mechanistic insights into this asymmetric yne-allylic substitution. The control experiment using **1a-Ph** and indole **2a** under the standard conditions in Scheme 2 did not lead to any product (Scheme 6a), demonstrating the importance of terminal alkyne in the reaction. This was further demonstrated by the use of D-labeled substrate **1a-d**, and the reaction afforded **3f** with D atom lost during the transformation (Scheme 6a). The relationship between the ee values of **L4** and the ee values of **5a** shows a proximal linear kinetic profile, indicating that a monomeric copper-allenylidene complex is probably involved in the reaction (Scheme 6b). Similarly, the reaction producing **8a** using ligand **L1** also gives a linear correlation, showing that a copper−ligand monomer complex possibly exists in the mechanism (Scheme 6b). Further kinetic studies revealed that the reaction was first-order on the Cu catalyst and zero-order on both **1a** and **2f** (Scheme 6c). This further confirms that the monocopper species is probably involved in the turnover-limiting step.

The postulated mechanism of this asymmetric yne-allylic substitution is described in Scheme 7. The yne-allylic carbonate **1a** first reacts with copper catalyst under base conditions to give alkynyl copper intermediate **Int-I**, which subsequently generates Cu-allenylidene intermediate **Int-II** with a tautomer of **Int-II'**. Then, an outer-sphere nucleophilic attack by electron-rich arenes occurs via remote enantiodiscrimination and finally generates products **5a** or **8a'**. An intramolecular annulation happens for **8a'** under copper catalysis and produces **8a** as the final product.

CONCLUSIONS

In summary, asymmetric copper-catalyzed yne-allylic substitution using electron-rich arenes has been developed for the first time, delivering a variety of 1,3-enynes bearing heterocyclic ring moieties and allenyl dihydronaphthofurans. The products contain a diverse set of valuable moieties, such as conjugated enyne, indole, indolizine, allene, and dihydrofuran, which are either key intermediates in organic synthesis or important substructures in bioactive or natural substances. The synthetic value of this protocol has been demonstrated in a series of further transformations, and mechanistic studies show that the ligands play a different role in delivering the products. The work provides a new choice of achieving remote stereocontrol via transition-metal catalysis and enriches the toolbox for bioactive molecule discovery. Further investigation of the novel application of asymmetric yne-allylic substitution is ongoing in our group.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.3c06146.

Experimental procedures, spectral data, and crystallographic data (PDF)
Compound **3k** checkCIF/PLATON report (PDF)
Compound **8b** checkCIF/PLATON report (PDF)

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Author Contributions

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Notes

The authors declare no competing financial interest.

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